Exploratory Data Analysis Report

Patients, clinical, operations, finance and patient experience

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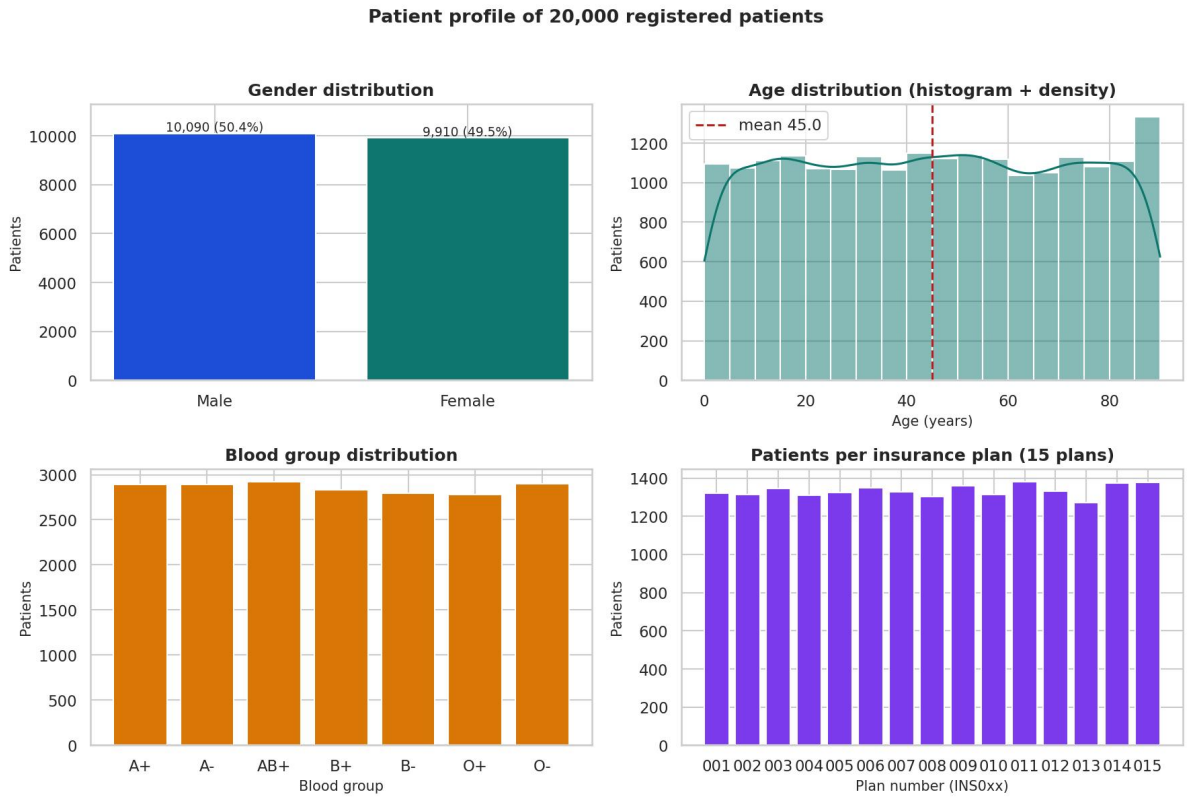
Scope and data caveats

This report is one of four companion reports for the Data Science track (Data Quality, Exploratory Data Analysis, Statistical Analysis, Predictive Analytics), all produced by the executed notebook HealthConnect_DataScience_Notebook.ipynb. Five tables named in the brief were not supplied (Health Facilities master, Laboratory Tests, Prescriptions, Admissions, Referrals); only 5,000 of 60,000 bills, diagnoses and payments link to an appointment; and the data behave like simulated data. Analyses that depend on missing tables use labelled proxies or are marked not computable.

Exploratory data analysis

Patients

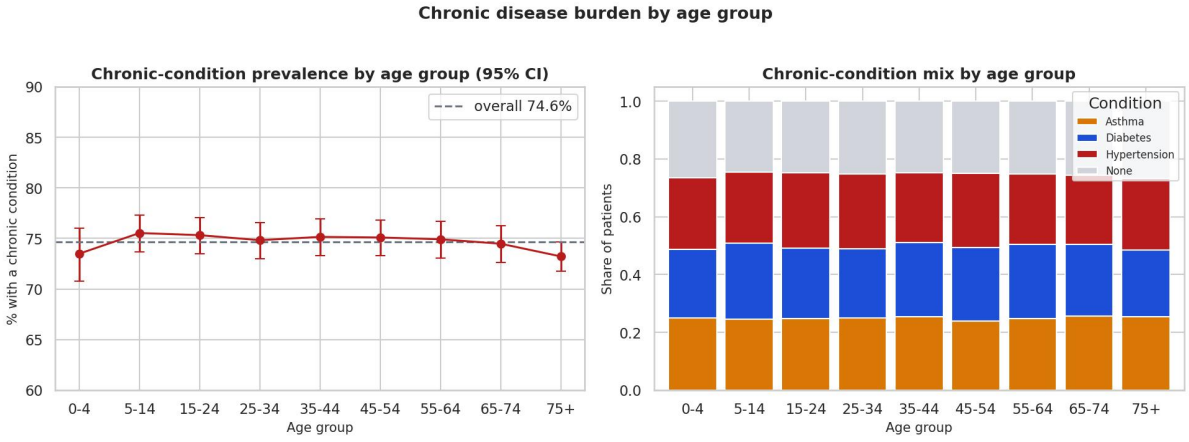
There are 20,000 registered patients, all marked Active; 4,443 (22.2%) appear in the appointment extract. The register is 49.5% female, mean age 45.0, 74.6% with a chronic condition (asthma 25.0%, hypertension 24.9%, diabetes 24.7%). Every patient holds one of 15 insurance plan IDs, so there is no uninsured group.



Interpretation: The register is almost perfectly balanced by sex (49.5% female) and spans all ages evenly (mean 45, no young-adult or elderly skew); the seven blood groups and 15 insurance plans are also near-uniform. Real populations are age-skewed and O+ is normally the most common blood group, while here only 7 of the 8 ABO/Rh groups appear (AB- is absent), which strongly suggests simulated data.

Figure: Patient profile of 20,000 registered patients

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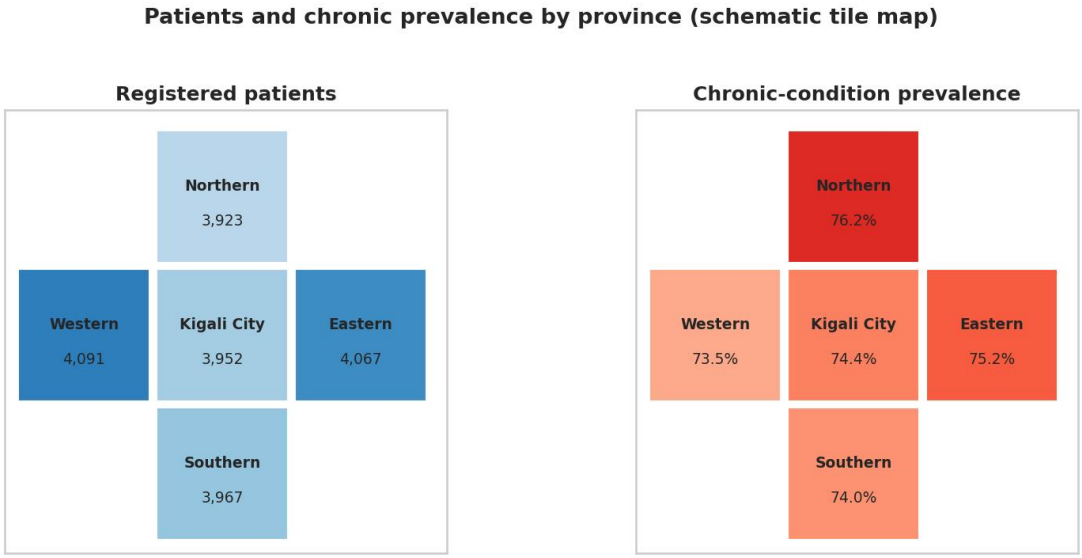


Interpretation: Prevalence is flat at ~75% in every age band (range 73-76%), and children carry the same burden as the elderly (1,005 under-18s are recorded with hypertension and 984 with diabetes). Biologically, chronic disease should rise sharply with age, so age cannot be used as a risk signal in this dataset.

Figure: Chronic disease burden by age group

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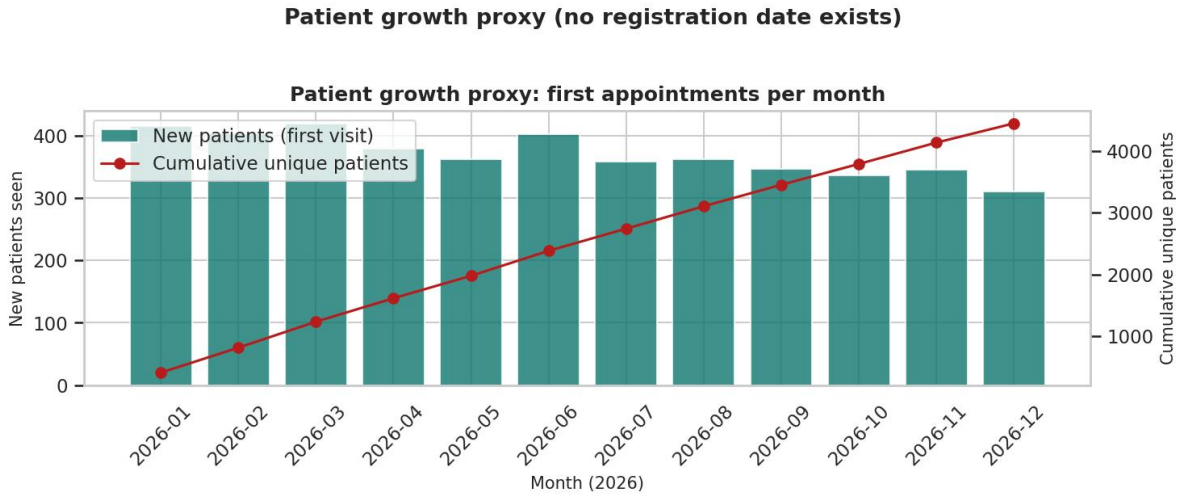
Biologically, chronic disease should rise sharply with age, so age cannot be used as a risk signal in this dataset.



Interpretation: Registered patients are spread evenly (3,923-4,091 per province) and chronic prevalence differs by under 2 points between provinces. A schematic tile map is used because district/sector/village names are placeholders and no boundary geometry exists, so a true geographic map is not possible.

Figure: Patients and chronic prevalence by province (schematic tile map)

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Interpretation: First-time patients fall from 416 in January to 311 in December (slope -8.7 a month, p<0.001). This is a counting artifact, not shrinking demand: as the year goes on more patients have already been seen, so fewer appointments can be a 'first' visit, while total appointments stay flat (~417 a month). 4,443 of 20,000 registered patients (22.2%) appear in the extract; true patient growth needs a registration date.

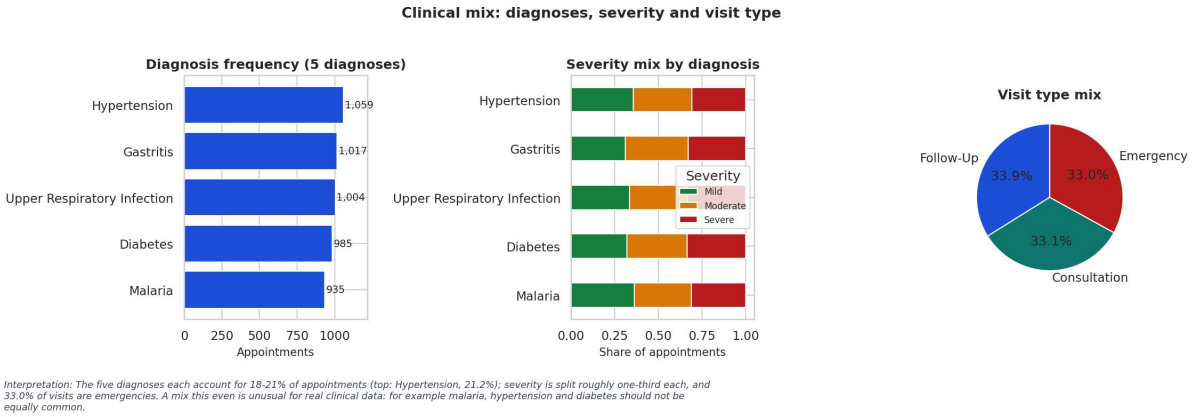
Figure: Patient growth proxy (no registration date exists)

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Clinical

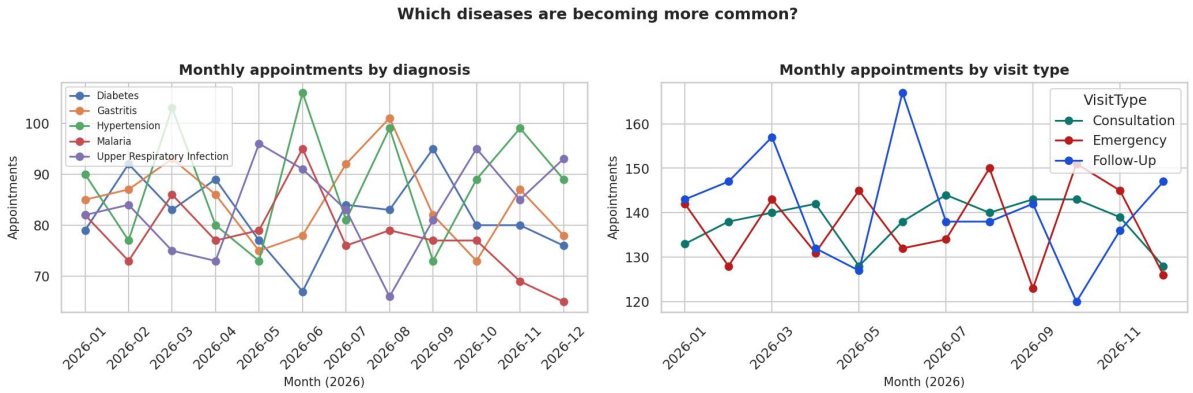
Only five diagnoses exist, so a "Top 20" list reduces to five. Visit type is the only acuity indicator available.



Interpretation: The five diagnoses each account for 18-21% of appointments (top: Hypertension, 21.2%); severity is split roughly one-third each, and 33.0% of visits are emergencies. A mix this even is unusual for real clinical data: for example malaria, hypertension and diabetes should not be equally common.

Figure: Clinical mix: diagnoses, severity and visit type

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Interpretation: Every diagnosis fluctuates around ~83 cases a month with no significant upward or downward trend (smallest Holm-adjusted p = 0.49), so no disease can be called 'increasing' on this evidence; month-to-month moves are noise.

Figure: Which diseases are becoming more common?

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Operations

Resource	Count
Doctors	350
Nurses	700

Resource	Count
Pharmacists	50
Laboratory scientists	40
Radiographers	30
Administrative staff	80
Rooms / beds	400 / 2,000 (all beds "Available"; occupancy not computable)
Equipment items	200 (10 per facility; availability not computable)

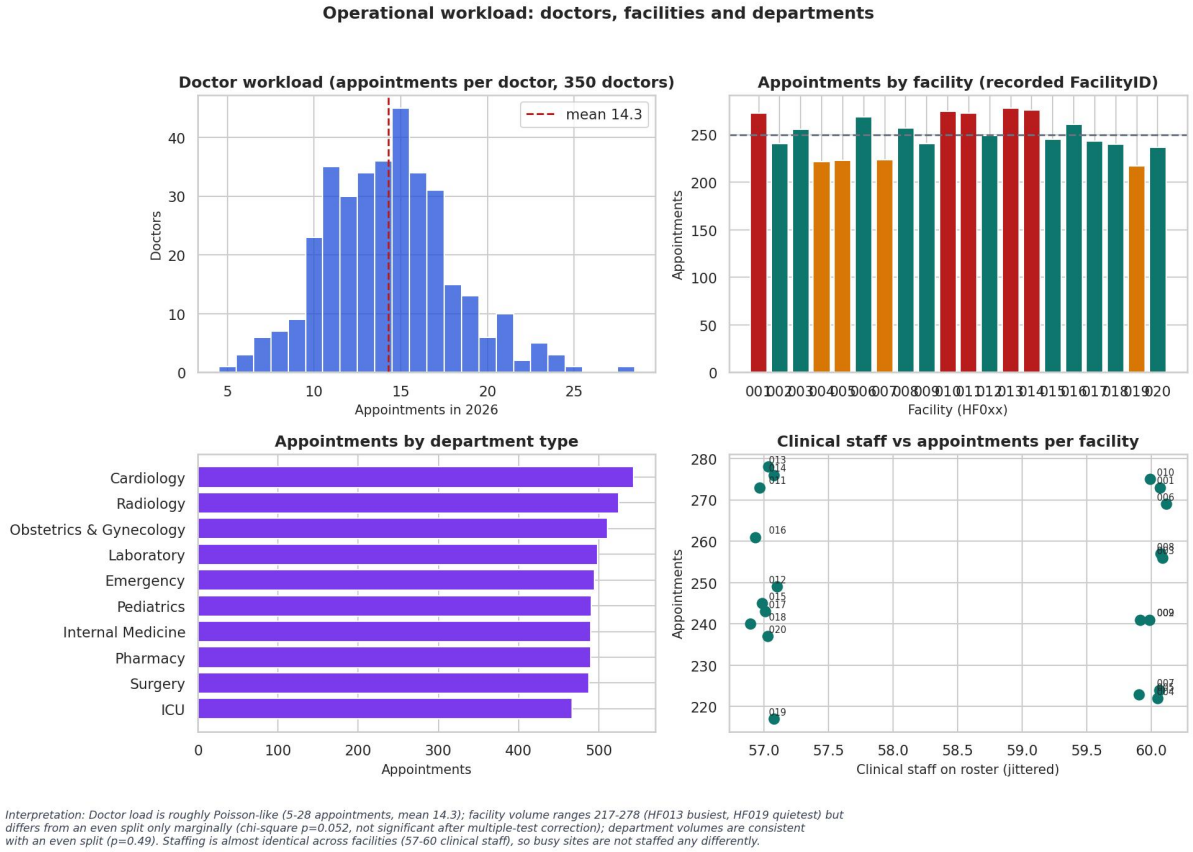


Figure: Operational workload: doctors, facilities and departments

Doctor load is roughly Poisson-like (5-28 appointments, mean 14.3); facility volume ranges 217-278 (HF013 busiest, HF019 quietest) but differs from an even split only marginally (chi-square p=0.052, not significant after multiple-test correction); department volumes are consistent with an even split (p=0.49). Staffing is almost identical across facilities (57-60 clinical staff), so busy sites are not staffed any differently.

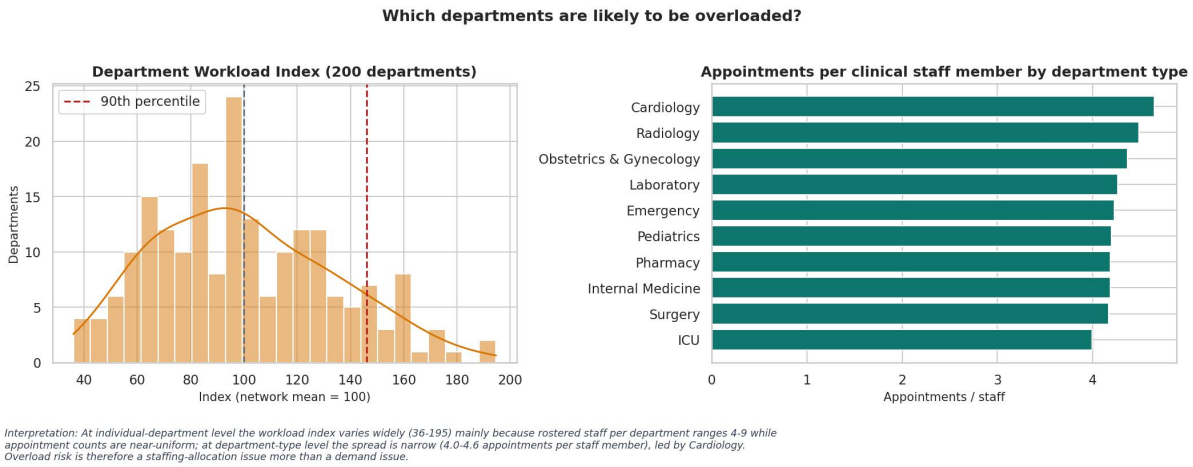
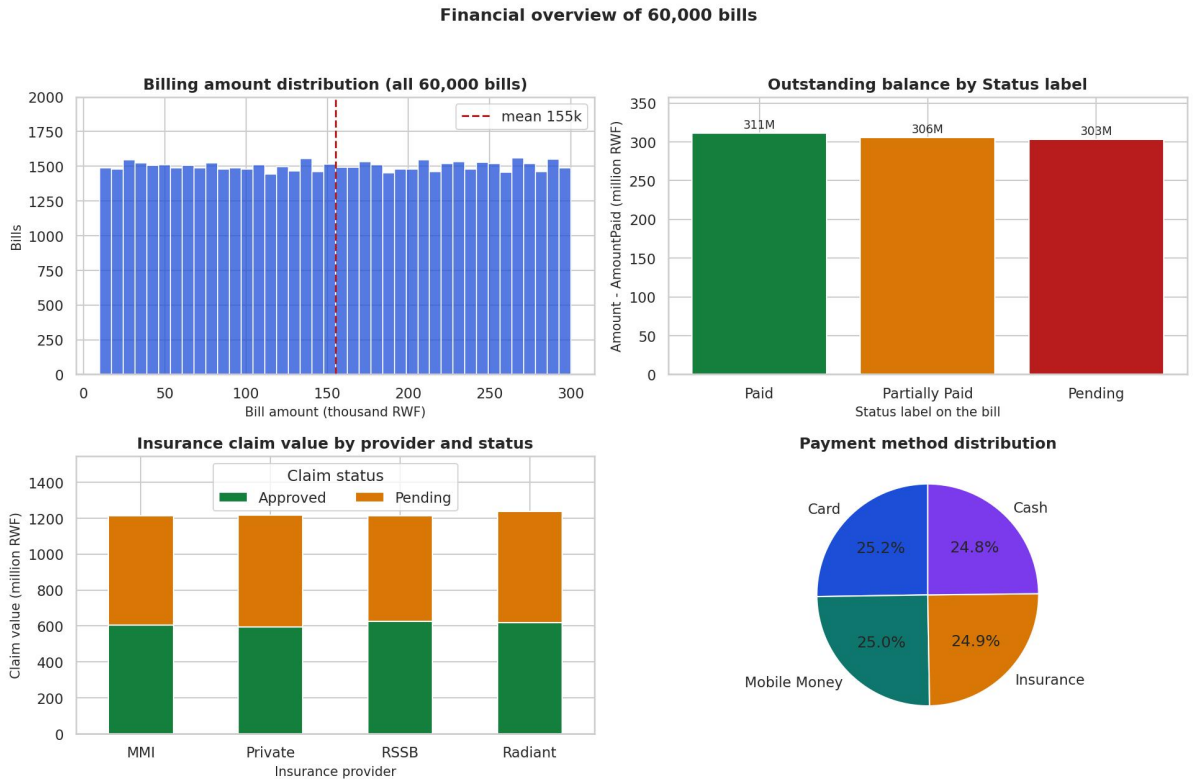


Figure: Which departments are likely to be overloaded?

At individual-department level the workload index varies widely (36-195) mainly because rostered staff per department ranges 4-9 while appointment counts are near-uniform; at department-type level the spread is narrow (4.0-4.6 appointments per staff member), led by Cardiology. Overload risk is therefore a staffing-allocation issue more than a demand issue.

Financial

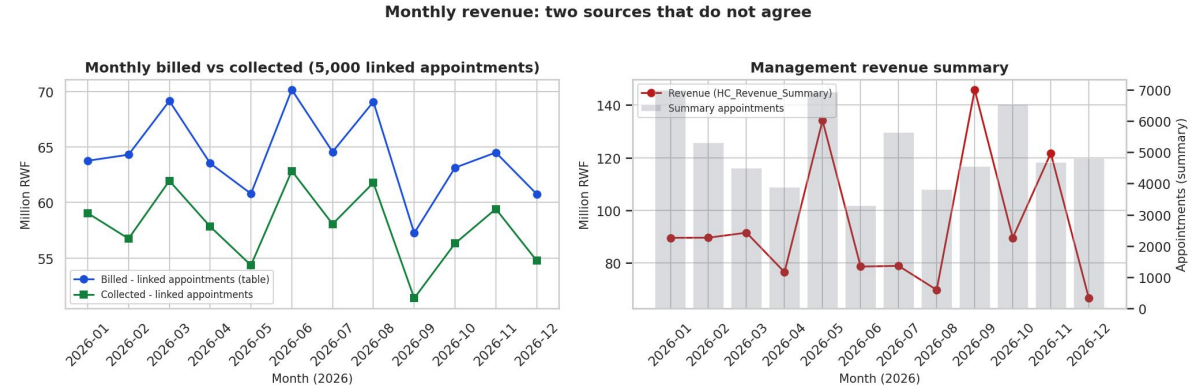
Billed value is 9.31bn RWF across 60,000 bills (mean 155,157); 8.39bn was collected, leaving 0.92bn (9.9%) outstanding. Of 42,036 insurance claims, 21,028 (50.0%, 2.44bn RWF) are pending; none are recorded as rejected. Payment methods are evenly split four ways.



Interpretation: Bills are uniformly spread between 10k and 300k RWF (mean 155k). 80.2% of bills are fully paid and the rest are underpaid by 0-50%, leaving 0.92bn RWF (9.9% of billings) outstanding, and the shortfall is the same whether a bill is labelled Paid, Partially Paid or Pending, so the status label carries no information. 50.0% of insurance claims are still pending.

Figure: Financial overview of 60,000 bills

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Interpretation: Billed revenue from linked appointments is stable (~64M RWF a month, range 57-70M), whereas the management summary swings from 67M to 146M with peaks in May, September and November. Summary revenue is unrelated to summary appointment counts (Spearman rho=0.33, p=0.30), so the swings are not volume-driven and need finance to explain them.

Figure: Monthly revenue: two sources that do not agree

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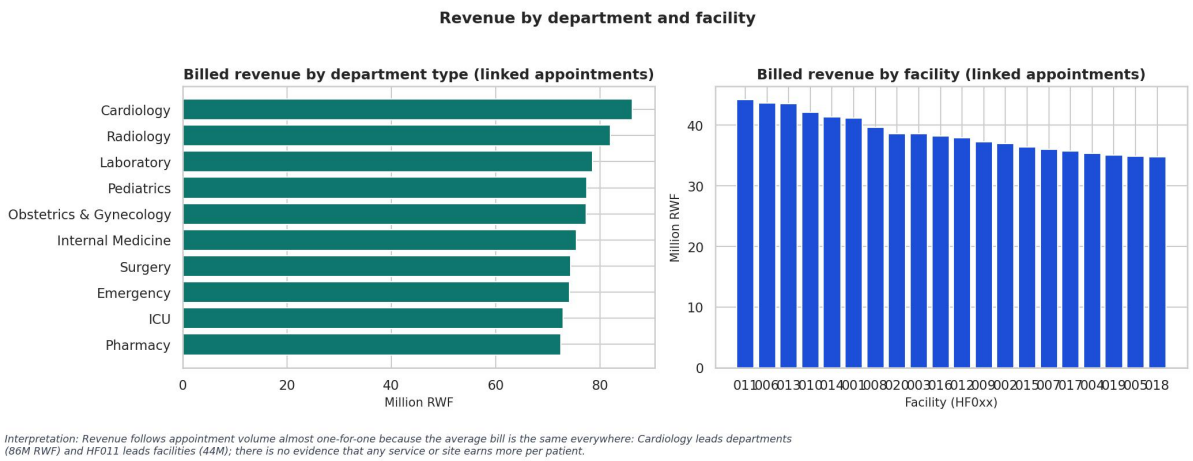
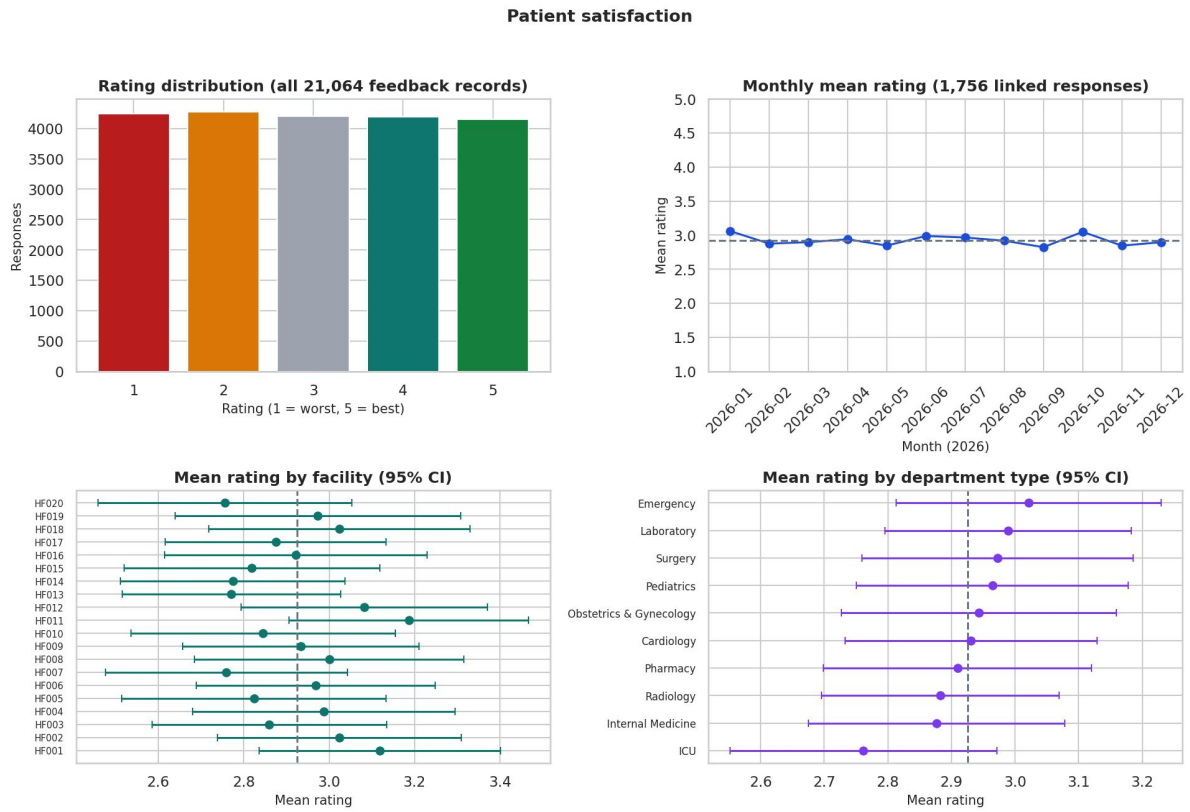


Figure: Revenue by department and facility

Revenue follows appointment volume almost one-for-one because the average bill is the same everywhere: Cardiology leads departments (86M RWF) and HF011 leads facilities (44M); there is no evidence that any service or site earns more per patient.

Patient experience

Across 21,064 ratings the mean is 2.99; 40.4% are 1-2 and 39.6% are 4-5. Only 1,756 link to an appointment, so facility and department comparisons use those. 30.1% of ratings contradict their text comment.



Interpretation: Ratings are evenly spread across 1-5 (mean 2.99; 40.4% rate 1-2, 39.6% rate 4-5). Facility means (2.76-3.19) and department means (2.76-3.02) differ, but all confidence intervals overlap and the tests in section 6.3 show the gaps are within chance. 30.1% of responses have a text comment that contradicts the star rating.

Figure: Patient satisfaction

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